ContextLinkr — An R Package for Creating Data Linkages with Contextual Variables

Todd Burus, MAS (under the supervision of Pamela C. Hull, PhD)

Scientific Rationale. Epidemiologic research involving individual-level data allows researchers to directly measure the association between individual characteristics/behaviors and health outcomes. Understanding the context in which an individual lives and how that impacts their health behaviors and outcomes is also an important consideration, as highlighted by McLeroy's Social-Ecological Model.[1–4] Performing research that incorporates information at multiple levels requires having both a sufficient amount of contextual data to analyze and the ability to connect it back to the individual records.[5] Unfortunately, much of the commonly used contextual data in the United States exists across multiple sources, in multiple formats, at multiple geographic levels, and is updated at multiple frequencies, making it time consuming to prepare. Moreover, differences in how patient location is collected can present difficulties in connecting them to their appropriate geographic contexts. Data cleaning and geospatial methods available within the R programming language can be leveraged to address the issue of providing geographic references for location data, though linkages with appropriate contextual variables would still need to be made. We have previously developed a platform called Cancer InFocus that automates the gathering and cleaning of publicly-available county- and Census tract-level data from over a dozen commonly used sources.[6,7] This data is updated monthly and available through https://cancerinfocus.org. The primary use of the Cancer InFocus platform to date has been in facilitating the creation of cancer surveillance visualization dashboards among US cancer centers. For this project, we propose to address the existing challenges by combining the data gathering power of the Cancer InFocus platform alongside the geospatial capabilities of the R programming language to develop an open-source R package for linking individual-level data with an extensive collection of contextual variables, making it easier to prepare datasets for multi-level analysis of health outcomes.

Innovation. The need for solutions to link individual-level data with contextual variables is growing throughout health outcomes research, driven in part by the widespread use of electronic health records and the existence of tissue repositories like The Cancer Genome Atlas. Though such solutions are available through certain software applications, these generally require paid subscriptions to access the software and/or highly specialized knowledge to be able to use them. To our knowledge, there does not currently exist a no-cost solution written in an easily accessible programming language or containing the breadth of data available through the Cancer InFocus platform. By creating an open-source R package that incorporates all of this, we would be putting a free, thorough, and reproducible way to make data linkages at the fingertips of biostatisticians and health researchers. Plus, by modularizing the steps in the process into individual functions within the package, users can easily apply different steps in computing environments with different levels of security as necessitated by their data. Moreover, this modularization also allows for easy use in applications beyond just data linkages with individuals. For instance, users could geocode and obtain contextual variables for a set of clinic sites they are interested in using for an intervention study, or simply pull down a set of contextual variables for a desired county or state. Future work could also be done on the package to develop functions for performing multi-level analysis as a complement to the data linkage utility.

Methodology. To achieve the goal of this project we will first write functions in the R programming language that execute various steps in the process of connecting individual-level data to contextual variables. These include the following functions (**Figure 1**):

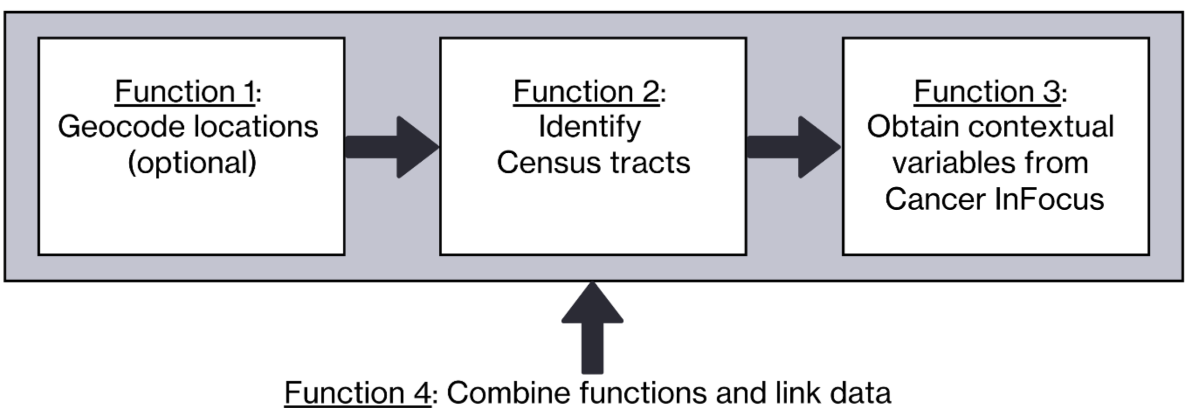
- **Function 1: Geocode individual-level data from available location information**. This function will take an individual-level data set with location information as an input and provide geocoded coordinates. Ideally, the initial dataset should include information about street address, city, state and ZIP code for each individual to obtain accurate geolocations, though coordinates can be obtained as long as either city and state or ZIP code alone are provided. Location information will be identified for each individual and passed to the US Census Geocoder API, followed by the OpenStreetMap API if necessary. Location coordinates will be attached to the original dataset and returned as output. NOTE: This function should only be used if the user has permission to apply open-source geocoding methods to the individual-level dataset. Otherwise, the user will need to perform geocoding in an approved, secure environment and skip to Function 2.

- **Function 2: Identify Census tracts containing individuals from the individual-level dataset.** This function will take a dataset with coordinate information for each entry as an input and provide the corresponding Census tracts containing them. This function will identify the coordinate locations for each entry, download a Census tract cartographic boundary file from the US Census Bureau, and perform a spatial join between the coordinates and the boundary file to identify overlaps. The output will consist of a list of Census tracts containing coordinates from the original dataset.

- **Function 3: Download contextual variables from Cancer InFocus filtered to a set of Census tracts**. This function will take a list of Census tracts and desired variable categories as an input and return a dataset consisting of contextual variables for each Census tract as the output. Desired categories of variables will be downloaded at the county- and/or Census tract-levels from the Cancer InFocus US dataset, filtered to the given Census tracts, and joined together to create a single contextual variable dataset.

- **Function 4: Join individual-level dataset to contextual dataset**. The function will be a wrapper function that takes an individual-level dataset with either (1) location information, or (2) coordinate information as an input and applies the previously described functions consecutively to link contextual variables to each entry. If only location information is provided, this function will apply functions 1-3 from above. If coordinates are already provided in the input dataset, this function will only apply functions 2-3.

Figure 1. Diagram of Core Functions. An illustrated guide to the relationship between functions used to link individual-level data with contextual variables in the ContextLinkr package. Functions 1-3 can stand alone or be used in series with the Function 4 wrapper function.



After the necessary functions to complete the desired process are developed and tested, they will be compiled into a cohesive, open-source R package and loaded into a public GitHub repository where users anywhere can access it for free. Though this package is being developed to facilitate the process of linking individual-level data to contextual variables, modularizing it into individual functions (1-3 above) will allow users with other use cases or needs to apply only parts of the process. For instance, if a user only wants to obtain contextual variables for a specific geography (such as a city or state), they can use Function 3 in isolation to generate the desired output. Additional helper functions will also be developed and added to the package to assist users in working with the data. Examples of possible helper functions include a function to pivot from wide to long datasets, a function to provide summary statistics on contextual variables obtained, and a function to map geocoded data. Sample data for testing and becoming acquainted with the package features will also be placed in the repository. Documentation will be developed and provided along with source code according to community standards for building an R package. An open-source license will be included and the package will be registered in Zenodo to obtain a digital object identifier.

Once the R package has been finalized, information about the package will be disseminated through various channels such as email listservs and newsletters. We will also prepare a manuscript about the package's development and use cases. Metrics on downloads, commits (e.g. collaborative input on package development), issue reporting, and issue resolution will be tracked through the package repository.

Timeline. This project will run from July 1, 2025 until June 30, 2026. The first six months will be used to develop and test the proposed data linkage functions along with additional helper functions for the R package. Months seven through nine will be used to prepare documentation, obtain an open-source license and digital object identifier, and deploy the package on GitHub. The final three months will be used to disseminate the package, troubleshoot any issues that arise during early adoption, and prepare a manuscript.

Institutional Review Board Status. The proposed project was deemed not human research by the University of Kentucky Institutional Review Board.

Budget. The only cost-incurring item for this project is salary effort to ensure dedicated time for the applicant to work on the project. The applicant works as a Data Scientist on the team of Dr. Pamela Hull (Behavioral Science faculty sponsor) in the Community Impact Office of Markey Cancer Center and is also a student in the Epidemiology & Biostatistics PhD program (Dr. Hull is on the applicant's dissertation committee). The applicant’s current base salary is $82,002. The applicant projects that this project will take 4% of their annualized effort over the 12-month project timeline (6% for months 1-6 and 2% for months 7-12). Factoring in a fringe rate of 19.237% and annual health insurance cost of $11,832, this works out to a total budget request of $4,384 for this project.

Benefit to Applicant. The proposed project will benefit the applicant by allowing time to develop a use case for the Cancer InFocus platform that he and the faculty sponsor have previously developed. The Cancer InFocus platform has already demonstrated success at being utilized to create cancer surveillance dashboards, but additional uses of this platform for facilitating innovative cancer research have not been explored. Building out additional use cases for Cancer InFocus will further demonstrate its utility and bolster the justification for funding its continued development and maintenance through external grant funding, including an R03 submission that he and the faculty sponsor are currently preparing. This specific use case could be utilized on studies involving patient-level data from the Kentucky Cancer Registry that the applicant is in the process of starting. Moreover, the proposed project will allow the applicant to gain experience in developing an open source R package. Receiving funding for this pilot would also give the applicant experience as a principal investigator on a funded project, helping him build a track record for future proposals for external funding after he completes his PhD and transitions to a faculty position. The applicant has not received previous pilot funding through the Department of Behavioral Sciences.

References

1. McLeroy KR, Bibeau D, Steckler A, Glanz K. An Ecological Perspective on Health Promotion Programs. *Health Education Quarterly*. 1988;15(4):351-377. doi:10.1177/109019818801500401

2. Diez-Roux AV. Bringing context back into epidemiology: variables and fallacies in multilevel analysis. *Am J Public Health*. 1998;88(2):216-222. doi:10.2105/ajph.88.2.216

3. Goel N, Hernandez A, Stoler J. Geospatial Genomics: Operationalizing Multilevel Conceptual Frameworks of Cancer Health Outcomes to Account for Gene-Person-Place Relationships. *JCO*. 2025;43(13):1534-1538. doi:10.1200/JCO.24.00751

4. Goel N, Hernandez A, Cole SW. Social Genomic Determinants of Health: Understanding the Molecular Pathways by Which Neighborhood Disadvantage Affects Cancer Outcomes. *JCO*. 2024;42(30):3618-3627. doi:10.1200/JCO.23.02780

5. Galvez-Hernandez P, Gonzalez-Viana A, Paz LG de, Shankardass K, Muntaner C. Generating Contextual Variables From Web-Based Data for Health Research: Tutorial on Web Scraping, Text Mining, and Spatial Overlay Analysis. *JMIR PUBLIC HEALTH AND SURVEILLANCE*.

6. Burus JT, Park L, McAfee CR, Wilhite NP, Hull PC. Cancer InFocus: Tools for Cancer Center Catchment Area Geographic Data Collection and Visualization. *Cancer Epidemiology, Biomarkers & Prevention*. Published online May 24, 2023:OF1-OF5. doi:10.1158/1055-9965.EPI-22-1319

7. Burus T, McAfee CR, Wilhite NP, Hull PC. Measuring the impact of a catchment area surveillance tool on cancer center adopters. *Cancer*. 2025;131(2):e35710. doi:10.1002/cncr.35710

8. Kingsbury P, Abajian H, Abajian M, et al. SEnDAE: A resource for expanding research into social and environmental determinants of health. *Computer Methods and Programs in Biomedicine*. 2023;238:107542. doi:10.1016/j.cmpb.2023.107542